



Access to finance and the exchange rate elasticity of exports [☆]

Mi Dai ^a, Francesco Nucci ^{b,*}, Alberto F. Pozzolo ^c, Jianwei Xu ^a

^a Beijing Normal University, China

^b Sapienza University, Italy

^c Roma Tre University, Italy



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ABSTRACT

The recent empirical literature estimating the elasticity of exports to exchange rate fluctuations has shown that, while devaluations have in general a positive effect on exports, the size of it varies significantly depending on firm-specific characteristics. Using Chinese firm-level data, we show that exporting activities by more financially constrained firms are more sensitive to exchange rate changes than those by firms with a better ability to raise external capital. This finding is detected at both the intensive and extensive margin of exports. Consistent with the result on export volumes, we also document that the exchange rate pass-through to export prices denominated in the domestic currency is lower for firms facing stronger financial constraints. Moreover, we establish that the role of financial conditions in determining a differential effect of exchange rate on exports continues to be relevant when we allow for a variety of alternative features that may also affect the firm-level elasticity of exports to exchange rate. These are the intensity of use of imported inputs, labor productivity, the degree of price stickiness and firm size.

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1. Introduction

Estimating the elasticity of exports to exchange rate movements has a long tradition in international economics, and a large amount of empirical evidence is now available showing that, while devaluations have a positive effect on exports, the impact varies significantly depending on firm, sector and country characteristics. A parallel strand of literature has established that a firm's higher ability to access external finance has a positive impact on export activities.

The purpose of our paper is to investigate if differences across firms in their financial conditions contribute to explain the heterogeneous effects of exchange rate fluctuations on firm exports. Our analysis builds on two sets of findings in the international trade literature. First, the responsiveness of exports to exchange rate movements is shaped by firm-specific features. For example, building on the seminal contribution of Melitz (2003), Berman et al. (2012) document that exports of more productive firms exhibit a lower response to exchange rate changes.¹ Another mechanism deals with the role of imported intermediate inputs, because firms relying more heavily on them have a lower sensitivity of exports to exchange rate, as shown

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* Corresponding author.

E-mail address: francesco.nucci@uniroma1.it (F. Nucci).

¹ Their finding hinges on the lower price elasticity of demand for these firms that allows them to adjust the markups more extensively in the wake of a currency swing. A similar result is found by Li et al. (2015) using data on exports of Chinese firms.

by Greenaway et al. (2010). Second, financial market imperfections restrain the ability of a firm to participate in the export markets, affecting both its decision of entry (i.e., the extensive margin) and the volume of its exports (i.e., the intensive margin), as convincingly established by, for example, Manova (2013), Muuls (2015) and Chaney (2016). Indeed, export activities impose specific costs on firms, that typically need to be paid upfront, increasing their working capital requirements. These costs can be fixed (such as those due to custom and regulatory compliance, to the expenses for gathering information on the potential destination markets, or to the effort of establishing a foreign distribution network) or variable (such as transportation costs, the costs of maintaining the foreign distribution network, or those related to cross-border payment services), but they all cause exporting firms to depend more on external finance with respect to those engaged only in domestic transactions.

The heterogeneous ability of firms to finance their operations provides a channel through which exchange rate shocks affect differently their exporting activities. Several dimensions, however, may characterize this transmission mechanism and, in principle, it is not obvious in what direction a higher incidence of financial constraints affects the exchange rate elasticity of exports. Part of the literature has emphasized a causal link from exports to credit constraints. For example, the standard expansionary effect of an exchange rate depreciation on exports may induce an improvement in the firms' current and prospective cash flows, allowing them to pledge higher future earnings to their demand for external finance. In turn, this contributes to attenuate the financial constraints faced by the firm, increasing its ability to access foreign markets (see Dekle and Ryoo (2007)). Hence, for credit constrained firms, the positive effect on export of an exchange rate depreciation would be amplified. On the other hand, Chaney (2016) argues that an appreciation of the domestic currency increases the value of domestic assets in terms of foreign currency, inducing a relaxation of the firm's liquidity constraints.

Another strand of literature has taken instead a different perspective, considering borrowing constraints as given and studying how they may affect the firm-level exchange rate pass-through on export prices and, thereby, the exchange rate elasticity of export volumes. In a seminal contribution in this field, Strasser (2013) documents that, in the wake of an exchange rate depreciation, credit constrained firms tend to increase the export prices denominated in the domestic currency only to a limited extent. Kosaka (2014) reaches similar conclusions from a theoretical perspective, by developing a model centered on the relationship between financial constraints and the firm-level exchange rate pass-through.

We follow this second approach and analyze whether the heterogeneity across firms in the elasticity of exports to exchange rate depends on their ability to access finance. To guide our empirical analysis, we first present a simple theoretical model that, following Kosaka (2014), delivers two main testable predictions. First, a higher incidence of financial constraints amplifies the responsiveness of a firm's export activities to exchange rate swings. Second, the more severe are the firm's financial constraints, the lower is the degree of exchange rate pass-through on its export prices denominated in the domestic currency. Our empirical investigation is conducted on panel data of a large sample of Chinese manufacturing firms. We believe that the Chinese economy provides an ideal environment for our purposes, in light of three characteristics: (a) the presence of credit constraints that limit the ability of firms to raise external finance; (b) the high export propensity; and (c) the fluctuations in the external value of the Chinese currency.²

Our empirical findings document that the exporting activities of more financially constrained firms are indeed more sensitive to exchange rate oscillations, and this holds true at both the intensive and extensive margins of exports. Importantly, the magnitude of the differential effect is relevant in economic terms: for example, the estimated impact of exchange rate changes on export volumes for a firm at the 25th percentile of the distribution of one of our measures of availability of internal funds is more than fifty percent larger than that for a firm at the 75th percentile. To decipher the mechanism underlying this result, and consistent with the predictions of the model, we also establish that the degree of exchange rate pass-through to export prices denominated in the domestic currency is lower for firms facing higher financial constraints.

Moreover, we augment our baseline empirical framework that focuses on firms' access to finance and allow for a number of additional factors that may also affect the exchange rate elasticity of exports. They include the intensity of use of imported inputs, the productivity of labor, the degree of price stickiness and the size of the firm. In all cases, we show that, when our estimating framework is augmented to control for each of these features, our findings continue to hold. While we do not question the relevance of these alternative channels, controlling for them contributes to strengthen our causal interpretation based on financial constraints. In addition, we document that our main findings survive other extensions and robustness checks. The latter, in particular, deal with the following aspects: a) possible asymmetries in the effects on exports of appreciation vs. depreciation of the Chinese RMB, b) the use of firm-level exchange rates in the empirical specifications and c) an explicit control for the incidence of processing trade and the firm's processing status.

Our contribution relates closely to the paper by Strasser (2013) mentioned earlier, that also focuses on how financial constraints affect the responsiveness of export decisions to exchange rate changes. However, we complement his analysis along three dimensions. First, we rely on firm-level panel information on the actual export volumes and unit values, while the data in Strasser (2013) are entirely qualitative in nature.³ Second, differently from us, there is no information on export prices and inferring the exchange rate pass-through on export prices from estimates based on information on domestic price expectations

² Song et al. (2011) presents a thorough discussion of the financial imperfections in China, showing that, on average, firms rely heavily on internal funds to finance their activities, but the pattern is very far from uniform. As for the central role of exports, the average annual growth of China's real exports has been about 14 per cent over the 1990–2014 period and total trade relative to GDP in real terms rose from 31 to 76 per cent between 1996 to 2007, remaining roughly stable since then (see Alessandria et al. (2017)).

³ Concerning exports, they are based on a firm's self-assessment on a three-level scale of the direction of its exports over the subsequent months: (a) increase; (b) remain the same (c) decrease.

rests on some very strong assumptions (see the discussion of Strasser's paper in [Gopinath \(2013\)](#)). Third, in our empirical analysis we also control for a number of additional firm-level factors, different from financial constraints, that have been found in the literature to shape the relationship between exchange rate and export performance.

Our paper is also related to two recent papers by [Ma et al. \(2019\)](#) and [Li et al. \(2020\)](#), also studying the link between financial constraints and exports, but from which our analysis differs along several dimensions. From a theoretical perspective, we do not consider the impact of exchange rate fluctuations on credit constraints, as explicitly recognized in the background underpinning the analysis of [Ma et al. \(2019\)](#), who explicitly refer to the channel of transmission identified by [Chaney \(2016\)](#), or in the theoretical model of [Li et al. \(2020\)](#), who assume that the probability that a firm repays its loans decreases as the exchange rate depreciates. Instead, we assume that credit constraints are taken as given and study how their presence may affect the export decisions of firms. From an empirical perspective, we study a larger sample of firms, including exporters and non-exporters, and we focus on their aggregate export behavior, not on the bilateral level of exports towards each country. From an empirical perspective, this difference is crucial. Consider for example a firm that exports in several different countries and faces an appreciation of the bilateral exchange rate with respect to one specific country. While it will almost certainly reduce its exports towards that specific country, it will also very likely reorganize its presence in the other foreign markets, in order to equate its aggregate marginal costs with the marginal revenues in each country. By focusing on aggregate firm exports, our empirical framework is certainly less suited to study the effect of bilateral exchange rate fluctuations on bilateral exports, but it is more suited to assess how firm level characteristics affect the impact of average exchange rate fluctuations on the overall degree of internationalization of a firm. In this sense, our empirical framework is very close to that of [Berman et al. \(2012\)](#), who study the impact of productivity on the exchange rate elasticity of firm exports. For these reasons, our contribution complements the important results of [Ma et al. \(2019\)](#) and [Li et al. \(2020\)](#).

We organize the remainder of the paper as follows. Section 2 briefly summarizes the literature relevant for our analysis. Section 3 presents a simple model that discloses the main channels through which a firm's financial conditions affect the dependence of its exports on exchange rate oscillations. Section 4 discusses our measures of financial constraints and describes the data used in the empirical analysis. Section 5 reports the estimation results for export performance from the baseline specification and those for export prices. Section 6 presents a sensitivity analysis allowing for a number of alternative transmission channels while Section 7 focuses on other extensions and robustness checks. The last section concludes.

2. Related literature

Our analysis builds on two strands of literature on the determinants of firm exports. The first focuses on the incomplete pass-through of exchange rate fluctuations on export prices due to pricing-to-market by exporting firms. Indeed, if exporters kept their export prices fixed in terms of the foreign currency, then a depreciation of the exchange rate would have no impact on the volume of exports. A body of literature has shown that the degree of exchange-rate pass-through is very heterogeneous across firms, depending on a number of idiosyncratic characteristics. In a largely cited paper, [Berman et al. \(2012\)](#), consistently with the theoretical framework of [Melitz and Ottaviano \(2008\)](#), show that high-productivity firms react to a depreciation by increasing significantly more their markups and significantly less their export volumes, due to a lower price elasticity of demand for their products. Whilst this literature is growing fast, a consensus on the prevailing source of heterogeneity across firms in the degree of exchange rate pass-through on export prices has not yet been reached.

This leads us to the second strand of literature related to our analysis, that on credit constraints and exports. Building on the seminal contribution of [Kletzer and Bardhan \(1987\)](#), a number of authors have studied the link between finance and exports. From a theoretical perspective, the typical analytical framework is an extension of the [Melitz \(2003\)](#) model of international trade with heterogeneous firms, in which the degree of severity of financial constraints provides an additional source of firm's specificity. A common prediction of the models of [Manova \(2013\)](#), [Feenstra et al. \(2014\)](#) and [Chaney \(2016\)](#) is that financially constrained firms need to achieve a higher level of productivity than unconstrained firms to be able to export. This is because, in addition to cover the fixed and variable costs to access foreign markets, exporters must also bear the burden of the higher costs of the external finance that is necessary to pay ex ante such costs. From an empirical perspective, starting from the earlier contribution of [Beck \(2002\)](#), a large amount of evidence has been produced that confirms this theoretical prediction. In a seminal paper that follows the identification structure first proposed by [Rajan and Zingales \(1998\)](#), [Manova \(2013\)](#) shows that firms in sectors that are more dependent on external finance are more likely to export if they are based in countries with a higher level of financial development. Many papers use instead firm-level data from a single country ([Greenaway et al. \(2007\)](#), [Feenstra et al. \(2014\)](#), [Manova et al. \(2015\)](#) and [Minetti and Zhu \(2011\)](#)) or from more countries ([Berman and Hericourt \(2010\)](#), [Pietrovento and Pozzolo \(2019\)](#)), reaching similar conclusions.

Against this background, a factor external to the firm, such as exchange rate swings, is likely to affect export decisions differentially across firms. While the available evidence points to a negative and significant effect of financial constraints on exports, not much has yet been said about how differences across firms in the ability to access finance shapes the responsiveness of export behavior to exchange rate movements.

[Dekle and Ryoo \(2007\)](#) and [Guillou and Schiavo \(2014\)](#) make the hypothesis that an unexpected exchange rate depreciation not only increases the export competitiveness of a firm, but it can also affect its cash-flows, and therefore its ability to finance the productive activities. This singles out a channel through which firms' decisions are affected by exchange rate swings, whose sign depends on the assumed impact of a devaluation on the cash-flows. In particular, [Dekle and Ryoo \(2007\)](#) fit a structural model on data of Japanese firms and show that the export elasticities are larger in those industries

in which an exchange rate depreciation increases the firms' cash-flows. [Guillou and Schiavo \(2014\)](#) focus instead on the impact of exchange rate fluctuations on profits and conduct an empirical analysis on a panel of French firms. They find that profit sensitivity to exchange rate fluctuations increases (decreases) with the level of liquidity in firms belonging to those industries where a depreciation induces an increase (decrease) of liquidity. According to [Chaney \(2016\)](#), in the wake of an appreciation, financially constrained firms can more easily afford to pay the fixed costs associated to entry in the export market because the increase in the value of domestic assets in terms of foreign currency induces a mitigation of the firms' financial restraints.⁴ As discussed earlier, [Ma et al. \(2019\)](#) and [Li et al. \(2020\)](#) study a large sample of Chinese exporters, finding that the presence of financial constraints dampens the impact of the depreciation of the bilateral exchange rate on the exports to that country, reducing export value to the existing destination market, the number of products exported and the probability of entering a new destination market.

Conversely, [Strasser \(2013\)](#) and [Kosaka \(2014\)](#) focus on a different channel of transmission, studying how credit constraints affect the pass-through of exchange rate variations on price setting. [Strasser \(2013\)](#) argues that credit constrained firms have fewer margins to adopt pricing-to-market strategies and absorb exchange rate swings into the markups that they apply to their products. He documents that credit constrained firms pass through exchange rate variations to foreign prices at about twice the rate at which the unconstrained firms do and that these firms exhibit a stronger response of exports to exchange rate changes.

3. A simple theoretical framework

Preliminary to the empirical analysis, we present in this section a simple model where the impact of exchange rate changes on export activities depends on how the credit constraints faced by the firm affect its ability to pay the fixed and variable costs of exporting. A crucial assumption of our model is that credit constraints are assumed to be exogenous and independent of the level of the exchange rate. This allows us to highlight the specific transmission mechanism which we aim to uncover in the empirical analysis.

As in [Melitz and Ottaviano \(2008\)](#), we assume that each firm produces one variety from a continuum of differentiated products and sells it in a monopolistically competitive market. Each firm maximizes the profits that it earns by selling abroad and, for simplicity, we make the hypothesis that the firm only produces for the export market. Assuming preferences over a set of differentiated products as in [Melitz and Ottaviano \(2008\)](#), each firm i faces a linear inverse demand function for its product:

$$p_i = d - f q_i \quad (1)$$

where q_i is the quantity of variety i , p_i is its price expressed in the currency of the market where the good is sold, and d and $f > 0$ are constant parameters.⁵

The cost of producing one unit of output is $c a_i$, where c is the price of a cost-minimizing bundle of inputs and a_i is drawn from a common distribution $G(a)$ with support on $[0, a_M]$ and is the inverse of firm i 's productivity level, $1/a_i$. As in [Manova \(2013\)](#), we assume that, to sell its product in the foreign markets, the firm must also pay up-front both a fixed cost, F_i , and a variable cost, that takes the standard iceberg form, $\tau > 1$. The firm can pay these up-front costs using its available liquid assets, L_i , but if they are insufficient it must recur to external sources. The latter, however, are more costly than the internally generated funds because of financial market imperfections and a premium, $\phi_i > 0$, has to be paid with respect to the cost of internal funds (that, for simplicity, is assumed to be equal to zero). Such premium is firm-specific and reflect the severity of the credit constraints faced by the firm.⁶

Let e be the nominal exchange rate expressed as the amount of domestic currency to purchase one unit of the foreign currency, so that an increase in e is a depreciation. Under these assumptions, each firm maximizes the following profit function:

$$\Pi(e, p_i, q_i, F_i) = e p_i q_i - q_i \tau c a_i - F_i - (q_i \tau c a_i + F_i - L_i) \phi_i, \quad (2)$$

subject to the constraint given by the linear demand function (1).⁷

⁴ Along similar lines, [Desai et al. \(2008\)](#) argue that depreciations intensify financial constraints if firms have liabilities denominated in foreign currency, as is especially common in emerging markets. Moreover, they emphasize that liquidity constraints are often associated with short-term debt exposure and interest rates often rise after depreciations. Similarly, using country-level data [Berman and Berthou \(2009\)](#) document that the positive effect of a depreciation on exports tends to be weaker in a country where: a) the firms' propensity to borrow in foreign currencies is high and b) financial market imperfections are high.

⁵ The parameter $f > 0$ indicates the degree of product differentiation between the varieties, while d reflects the value of other preference parameters combined with the consumption level over all varieties, which we treat as constant.

⁶ Whilst simple, this way of modelling financial constraints is in line with the approach by [Kaplan and Zingales \(1997\)](#), who classify firms as credit constrained if they face a wedge between the internal and external costs of funds and a firm is considered more financially constrained if this wedge is higher. On the other hand, precisely as in [Kaplan and Zingales \(1997\)](#), we are agnostic on the sources of the wedge between the cost of internal and external funds (hidden information problems, agency problems or others).

⁷ To simplify the analysis, a number of possibly relevant aspects are abstracted in the model. For example, we do not allow for any effect of the exchange rate on the firm's expenditure for inputs, as we assume that firms rely exclusively on domestic, non-imported inputs. Different, for example, from [Feenstra et al. \(2014\)](#), we also make the hypothesis that both the firm-specific premium for external finance, ϕ_i and the level of liquidity, L_i are exogenous and they are not affected by the level of the exchange rate (see [Li et al. \(2020\)](#) for a different approach). Indeed, the model might be extended to consider different possibilities, for example allowing for imported inputs and assuming that part of the fixed and variables costs are denominated in foreign currency so that the firm's expenditure expressed in the domestic currency would increase if the latter depreciates, as in [Chaney \(2016\)](#).

The optimal level of production-export is a function of the exchange rate and the variables describing each firm's characteristics:

$$q_i = \frac{d}{2f} - \frac{\tau c a_i (1 + \phi_i)}{2ef} \quad (3)$$

The derivative of Eq. (3) with respect to the exchange rate, e , shows that, for an exporting firm, a depreciation always causes an increase in its levels of exports:

$$\frac{\partial q_i}{\partial e} = \frac{\tau c a_i (1 + \phi_i)}{2e^2 f} > 0. \quad (4)$$

As for the cross derivative, the derivative of Eq. (4) with respect to the external finance premium, ϕ_i , establishes that the size of the impact of an exchange rate depreciation on the volume of exports is an increasing function of the external finance premium:

$$\frac{\partial^2 q_i}{\partial e \partial \phi_i} = \frac{\tau c a_i}{2e^2 f} > 0. \quad (5)$$

We now insert the expression for q_i in Eq. (3) in the demand function (1) and obtain the following expression for the optimal price level, p_i (denominated in the foreign currency):

$$p_i = \frac{1}{2} \left[d + \frac{\tau c a_i (1 + \phi_i)}{e} \right] \quad (6)$$

Since a higher external finance premium, ϕ_i , causes the marginal costs to be higher, firms with more severe financial constraints tend to set higher prices and therefore, with a linear demand function, to face a higher price elasticity of demand.⁸ Thus, with markup values endogenously determined in the model, an exchange rate depreciation allows firms to increase their markups but the credit constrained firms do so only to a limited extent because they have fewer margins to adjust their profit margins. To delve deeper into the mechanisms underlying these predictions, we also focus on the role of the firm's financial constraints in shaping the degree of exchange rate pass-through to export prices. After differentiating Eq. (6) with respect to exchange rate, e , the firm-specific exchange rate pass-through elasticity of export price, $\eta_{p_i, e}$, is characterized as follows:

$$\eta_{p_i, e} = \frac{\partial p_i}{\partial e} \frac{p_i}{e} = -\frac{\tau c a_i (1 + \phi_i)}{ed + \tau c a_i (1 + \phi_i)} < 0, \quad (7)$$

so that the following result obtains:

$$\frac{\partial |\eta_{p_i, e}|}{\partial \phi} = \frac{\tau c a_i e d}{[ed + \tau c a_i (1 + \phi_i)]^2} > 0. \quad (8)$$

This establishes that the higher is the excess cost of external finance, the higher is the absolute value of the exchange rate pass-through elasticity of export prices denominated in the foreign currency (and, symmetrically, the lower is the exchange rate pass-through elasticity of export prices denominated in the domestic currency).

The previous results focus on the marginal effect of exchange rate oscillations on exports for an exporting firm. However, for a firm to be exporting, two conditions need to be satisfied: (i) that the equilibrium level of production from Eq. (3) is positive, and (ii) that the associated level of profits from Eq. (2) is also positive. Following the logic of Melitz (2003), Melitz and Ottaviano (2008) and Manova (2013), these conditions can be studied by solving for the cutoff level of productivity $1/a_i^*$ ensuring that the two previous conditions are satisfied. From Eq. (3), positive production requires that $a_i < \frac{ed}{\tau c(1+\phi_i)}$. By inserting in expression (2) the profit-maximizing value of q_i from Eq. (3), the condition for positive profits can be written as:

$$\frac{[ed - \tau c a_i (1 + \phi_i)]^2}{4ef} > (1 + \phi_i) F_i - \phi_i L_i. \quad (9)$$

Solving this condition with respect to a , we single out the cut-off level, a_i^* , and characterize the values of the inverse of productivity that are compatible with the firm making positive profits:

$$a_i < \frac{ed}{\tau c(1 + \phi_i)} - \frac{2\sqrt{ef(1 + \phi_i)F_i - f\phi_i L_i}}{\tau c(1 + \phi_i)} = a_i^*; \quad (10)$$

to be defined, this condition requires that the argument of the square root in the right hand side of the expression is non-negative, that is $F_i \geq \frac{\phi_i}{1+\phi_i} L_i$. Expression (10) provides a more stringent condition than that required for positive production.

⁸ Clearly, this result would not be confirmed with isoelastic preferences, that deliver a constant optimal markup.

Thus, whenever productivity is sufficiently high to guarantee positive profits (i.e., condition (ii) is satisfied), the firm produces a positive amount of its good (i.e., condition (i) is also satisfied).

The first derivative of a_i^* with respect to the exchange rate, e , in (10) is:

$$\frac{\partial a_i^*}{\partial e} = \frac{ed - \sqrt{ef(1 + \phi_i)F_i - ef\phi_i L_i}}{e\tau c(1 + \phi_i)}. \quad (11)$$

Simple algebraic manipulations allow us to show that this expression is positive if:

$$F_i < \frac{\phi_i}{1 + \phi_i} L_i + \frac{ed^2}{f(1 + \phi_i)}. \quad (12)$$

For a sufficiently small amount of fixed costs, F_i , relative to the amount of liquidity, L_i (i.e., whenever condition (12) is satisfied), we obtain the parallel result of Eq. (4): an exchange rate depreciation leads to a reduction of the minimum level of productivity that is required for a firm to be an exporter.

Finally, in the Appendix we also show that there is an interval of values of F_i such that the impact of an exchange rate depreciation on the minimum required productivity is stronger for firms that face higher costs of external financing, ϕ_i , similarly to what Eq. (5) establishes for the intensive margin.

A number of testable implications are therefore drawn from our simple model, that can be summarized as follows. *Result 1*: an exchange rate depreciation determines an increase of exports (the intensive margin; Eq. (4)). *Result 2*: the size of the effect of a depreciation on exports is an increasing function of the tightness of the financial conditions faced by the firm (Eq. (5)). *Result 3*: the degree of exchange rate pass-through on export price expressed in the domestic currency is lower for firms with a larger excess cost of external finance (Eq. (8)). *Results 4 and 5*: under some conditions, the results 1 and 2 apply also for the extensive margin (Eqs. (11) and (15)). All these theoretical predictions lend themselves to the empirical scrutiny, to which we turn in the next sections.

4. The empirical framework

4.1. Measuring financial constraints

A first critical issue in our empirical investigation is that of measuring the extent of financial constraints. The firm's ability to access external finance is not directly observable and so is the wedge between the internal and external cost of funds employed in our simple theoretical model, in line with Kaplan and Zingales (1997). A great deal of effort has been exerted in the empirical literature to propose indirect measures at the firm level. To approximate the intrinsically unobservable degree of financial constraints faced by a firm, we rely on information drawn from firms' balance sheet and cash flow statement items.

In doing so, we follow Kaplan and Zingales (1997) again and characterize unconstrained or less constrained firms as those firms with relatively large amounts of liquid assets and net worth. Although this measurement approach is widely used in both the international economics and corporate finance literature, we are aware of its shortcomings in measuring the true dependence on external financing.⁹

In light of this, instead of using a single index only, we rely on a variety of alternative indicators, each capturing a different dimension of the firm's ability to access finance. These time-varying indicators are: a) the ratio of cash holdings to total assets, b) the ratio of net liquidity to total assets, and c) a measure of leverage obtained as the ratio of debt to total assets. Cash holdings are computed as: current assets net of inventories and accounts receivable (see Love (2003)), while, following Manova and Yu (2016), the amount of net liquid assets is computed as current assets minus current liabilities. High levels of the cash holdings and the net liquidity ratios suggest that the firm has sufficient margins for mobilizing internal funds to finance exporting activities. On the other hand, a relatively high exposure of the firm on the debt side suggests that the firm is likely to face some difficulties in further increasing its reliance on external funds. In particular, as argued convincingly by Berman and Hericourt (2010), a high ratio of debt over total assets is likely to indicate both the firm's lack of collateral and a discrepancy between the firm's demand for borrowing and the current capacity to borrow. For clarity of exposition, in the econometric analysis we take the inverse of the ratio of total debt over total assets to ensure that an increase in each of our indicators represent a decrease, a relaxation of financing constraints.

As for the concerns associated to using balance sheet ratios to approximate financial constraints, a critic might argue that a high liquidity-to-assets ratio, under some circumstances, may indicate the presence, rather than the absence, of credit constraints. The argument would be that the ratio is high if accounts receivables are high, possibly signalling a constrained firm dealing with customers unable to pay their bills. As for the cash-to-assets ratio, under specific circumstances a high ratio may reveal inadequate cash management practices and thereby financial constraints, with an interpretation of this indicator

⁹ Starting from the seminal contribution of Fazzari et al. (1988), a number of indexes of financial constraints have been constructed using linear combinations of observable firms' features which are likely to reflect their ability to raise capital, such as the payment of dividends, a credit rating or balance sheet's information pertaining to the financial conditions and characteristics of the firm (i.e., the extent of non-equity leverage, cash flow, intangible assets and cash holdings). Notable examples are the indexes of Kaplan and Zingales (1997), Whited and Wu (2006) and, more recently, the index developed by Hadlock and Pierce (2010).

that is opposite to the standard one. Finally, a high assets-to-debt ratio (the inverse of the leverage ratio) may signal that the firm excessively relies on equity financing because it is unable to issue debt. In general, we do not rule out the possibility that some of the previous cases occur in some firms. However, by relying on a number of different financial indexes and by estimating every empirical equation separately for each of them makes us more confident on the soundness of our findings and their interpretation.

On the other hand, other approaches to measuring financial frictions are either not exempt from criticism or less suitable for our investigation at the firm level. For example, with the [Rajan and Zingales \(1998\)](#)'s type of methodology, where an industry-specific measure of external dependence is interacted with regional measures of financial development, the large heterogeneity across firms in the extent of credit constraints would be washed out in the aggregation process for constructing the indicator.¹⁰

4.2. The data

The microeconomic data used in this paper is panel information at the firm level referring to the Chinese economy. The source of our firm data are the Annual Surveys of Manufacturing Firms conducted by the National Bureau of Statistics of China over the years from 2000 through 2006. The surveys cover all enterprises with annual sales of five million Renmimbi (that is, about 650,000 US dollars) or more. The dataset includes information from balance sheets, firms' profit and loss statements and cash flow statements. Overall, data on about 100 variables are collected, providing detailed information on, for example, revenues, employment, the capital stock and the type of ownership. The firms included in the Surveys cover approximately 98 per cent of total Chinese manufacturing exports as recorded in the aggregate trade data.

Following [Feenstra et al. \(2014\)](#) closely, we clean the data for mismeasuring and drop observations that report missing or negative values for any of the following variables: total sales, total employment, fixed capital, export value, intermediate inputs. We also drop observations if the export value exceeds that of total sales or if the share of foreign assets over total assets exceeds one. We include firms with at least eight employees.

We also supplement this firm-level information with customs data collected and maintained by the China's General Administration of Customs. These data are recorded monthly and provides information on the quantity and value of trade flows for each product and with each firm's trading partner. Before merging these data with the firm-level information from the Annual Manufacturing Surveys, we aggregate them at the firm and annual level.¹¹ We use customs data on exports for two additional purposes: first, to construct a firm-level indicator of import intensity, which allows us to disentangle the impact of financial constraints and that of imported intermediate inputs on the elasticity of exports to exchange rates. Second, to calculate the unit value of exports used to study how the degree of exchange rate pass-through to export prices varies across firms depending on their ability to have access to finance.

Merging the two datasets is not an obvious task, as each of them employs different firm identifiers: a nine-digit ID in the Manufacturing Survey and an eleven-digit ID in the customs data, with no elements in common. We merge the two datasets using information on the firm's name, telephone number and zip code.¹² The resulting sample comprises about 100,000 firms with export values accounting for 54 per cent of the total export value recorded for the firms of the manufacturing Survey.

Nominal bilateral exchange rates are drawn from the International Financial Statistics (IFS), and the price indexes to construct the real effective exchange rates at the 4-digit industry level are the consumer price index data available in Penn World Tables. In particular, following [Dai and Xu \(2013\)](#), the real effective exchange rate is constructed as an index that takes the value of one in 2000, our base-year.¹³

Table 1 summarizes the key variables. A non negligible variability across firms is detected for both exports and the indexes of access to finance. As it is customary with firm-level data, the distribution of total exports is characterized by a high degree of skewness, as shown by the large difference between the mean and the value at the 95th percentile: not surprisingly, few very large firms account for a large share of Chinese exports. Accordingly, also the value of the standard deviation is high, implying a coefficient of variation of more than 250. Since our sample includes firms with total sales worth as small as 5 millions of Renmimbi (about 600,000 US dollars in year 2000), the share of exporters is just above 15 per cent. This provides strong motivation for broadening the analysis to the extensive margin of exports. Exchange rate variability is also significant, both through time and across industries. Indeed, while the Renmimbi mainly pegged to the US dollar in the

¹⁰ This approach is used, for example, by [Manova \(2013\)](#). See [Farre-Mensa and Ljungqvist \(2016\)](#) for a critical view on indirect measures of financial constraints that are routinely used in empirical analyses.

¹¹ We have chosen to focus on data aggregated at the firm and year level, instead of using information on exports towards each single trade partner, for two reasons: first, some of the costs of accessing foreign markets are independent on the number of destinations that are reached, and therefore their impact should be modeled as a non-monotonic function of the accessed destinations. Second, allowing for all potential destination countries would have generated a non-negligible curse-of-dimensionality problem.

¹² See Dai and Xu (2017) for a detailed description of the merging process.

¹³ More precisely, the real effective exchange rate at the industry level is defined as $REER_{CN,s} = \prod_{j=1}^n (REER_{CN,j})^{\frac{trade_{CN,j,s}}{\sum_{j=1}^n trade_{CN,j,s}}} = \prod_{j=1}^n \left(\frac{XRAT_{CN,j}}{XRAT_{CN,j,2000}} \frac{P_{CN,s}}{P_{j,s}} \right)^{\frac{trade_{CN,j,s}}{\sum_{j=1}^n trade_{CN,j,s}}}$, where $REER_{CN,s}$ is the real effective exchange rate for industry s in China, $trade_{CN,j,s}$ is the value of bilateral trade for industry s between China and country j , $REER_{CN,j,s}$ is the bilateral real exchange rate of industry s between China and country j , $XRAT_{CN,j}$ is the bilateral nominal exchange rate between China and country j , $XRAT_{CN,j,2000}$ is the nominal exchange rate between China and country j in 2000, our base year, and $P_{CN,s}$ and $P_{j,s}$ are the price levels of industry s in China and country j , respectively.

Table 1
Descriptive statistics.

variable	mean	median	s.d.	5th percentile	95th percentile
Total exports	6.70e+07	0	1.75e+10	0	1.80e+5
Exporter (dummy)	0.155	0	0.362	0	1
Exchange rates (total)	1.008	1	0.158	0.831	1.210
Exchange rates (by year)	1.023	1	0.090	0.946	1.155
Exchange rates (by industry)	1.022	1	0.279	0.856	1.164
Net liquidity over total assets	0.053	0.058	0.296	-0.465	0.541
Cash over total assets	0.209	0.163	0.172	0.013	0.564
Leverage	0.598	0.600	4.234	0.102	0.999
Correlations					
	Liquidity ratio		Cash ratio		Leverage
Liquidity ratio	1.0000				
Cash ratio	0.2938		1.0000		
Leverage	-0.6203		-0.0367		1.0000

Notes: Total exports are measured at constant prices of 2000 and are expressed in renminbi; the dummy variable Exporter takes the value of one for firms that are exporting in year t and zero otherwise. Exchange rates are the real effective exchange rates of the renminbi calculated at the sector level, and statistics by year and industry are computed on year and industry averages, respectively. Net liquidity over total assets is the ratio of current assets minus current liabilities over total assets. Cash is computed as current assets minus inventories and accounts receivable and its value over that of total assets is considered; Leverage is the ratio of debt over total assets.

first part of our sample period, the value of the dollar fluctuated extensively against the currencies of other countries with which China had relevant trade relationships. Since the real effective exchange rate is normalized to be one in the base year, the coefficient of variation is very similar to the standard deviation, whose overall value is 16 per cent. Interestingly, while most of the variability of the real exchange rate is due to the oscillations in the underlying nominal effective exchange rate, the part due to price oscillations is not negligible, as shown by the lower coefficient of variation of the corresponding nominal effective exchange rate, whose overall value is 8 per cent. Taking the average across industries, the standard error of the real exchange rate is about 9 per cent, mainly due to consecutive depreciations during 2001 – 2005, followed by a slight appreciation during 2005 – 2006. Taking the average across years, the variation across industries is about 28 per cent. Moreover, real exchange rates exhibited a higher degree of variability through time in some specific industries. For example, machinery and paper manufacturing experienced an appreciation of around 10 per cent, while industries like transport equipment experienced a strong depreciation of 15 per cent (Dai and Xu (2013)). Finally, all the three indexes of access to finance show a high level of variability, as shown by the high values of their standard deviation. For each indicator, the small difference between the mean and the median suggests that the distribution of these characteristics does not exhibit a high degree of skewness.

Net liquidity and cash are shown to be positively correlated with each other, and are negatively correlated with leverage, as expected. However, the relatively low values of some bilateral correlations, – notably with cash – suggest that the three indexes are likely to capture complementary aspects of access to finance.

Finally, consistent with the existing literature, Panel (a) of Fig. 1 indicates that, no matter which indicator is used, firms with better access to finance tend to export more. Similarly, Panel (b) shows that firms with relatively low levels of net liquidity and cash and high leverage are more likely to export.

4.3. The baseline specification and the estimation method

Let us now turn to the empirical analysis and test on microeconomic data the predictions of our simple model. In our empirical framework the baseline equation features as dependent variable either the volume of firm's exports (the intensive margin) or the decision to participate to exporting activities (the extensive margin). In particular, the specification for estimating the differential impact of exchange rates fluctuations on exports depending on the firm's ability to have access to finance is the following:

$$\begin{aligned} \ln(\text{Export volume}_{it}) = & \beta_0 + \beta_1 \ln(\text{Exchange rate}_{jt}) + \beta_2 (\text{Access to Finance}_{it-1}) \\ & + \beta_3 \ln(\text{Exchange rate}_{jt}) \times (\text{Access to Finance}_{it-1}) \\ & + \beta_4 \ln(\text{Labor productivity}_{it-1}) + \gamma' Z_{it} + \lambda_i + \psi_t + u_{it}, \end{aligned} \quad (13)$$

where $\text{Export volume}_{it}$ is the value of exports at constant prices of firm i in year t and $\text{Exchange rate}_{jt}$ is the industry-specific real effective exchange rate at time t , with j being the industry to which firm i belongs. An increase of this variable indicates a depreciation of the Chinese currency. $\text{Access to Finance}_{it-1}$ is the indicator of the firm's ability to have access to external finance and measures the intensity of financial constraints faced by firm i in year $t - 1$. As discussed in the previous section, for measuring this variable, we rely on a variety of alternative indicators drawn from the firms' balance sheet and cash flow

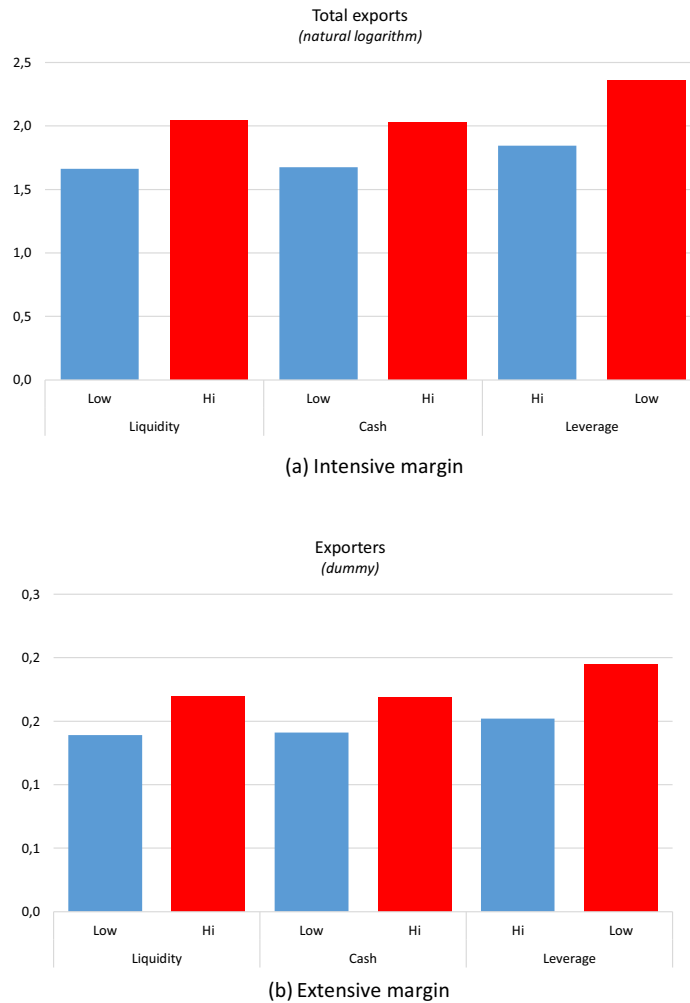


Fig. 1. Financial constraints and exports. Notes: Panel A presents the averages of log of total exports distinguishing (separately for each indicator of access to finance) between firms with a level of access to finance above and below the sample median. Panel B presents the average of the dummy variable taking the value of one if the firm is an exporter and zero otherwise, again distinguishing between firms on the basis of the value of the indicator of access to finance (below and above the median).

statement. These are: (1) the ratio of cash holdings to total assets, (2) the ratio of net liquid to total assets, (3) the ratio of total assets to debt. For the latter ratio, we take the inverse of the more traditional measure of leverage to ensure that, in all three cases, a higher value of the variable *Access to Finance*_{*t*-1} reflects a lower degree of financial constraints. These variables enter the specification with a one-period lag to mitigate the problems arising from their possible endogeneity.

In the estimating equations, in addition to including among the regressors the real exchange rate and the measure of financial constraints in isolation, we also insert the interaction among these two terms. This allows us to ascertain the differential effect of exchange rate on exports due to differences across firms in the access to external finance. The lagged log-level of labor productivity is also inserted in the specification. Finally, we control for Z_{it} , a vector of firm-specific variables, including dummy variables for the size of the firm, as measured by the number of employees, the industry and the type of ownership (distinguishing, among others, between domestic private and state-owned enterprises; see e.g. Song et al. (2011)).

The error term, u_{it} , is assumed to have finite moments with $E(u_{it})=E(u_{it}u_{is})=0$, for all $t \neq s$. ψ_t is a time effect captured by the inclusion of year dummy variables while λ_i is a firm-specific fixed effect. We assume that in Eq. (13) the vector of regressors, $\mathbf{w}_{it}$, is weakly exogenous in the sense that $E(\mathbf{w}_{it}u_{is})=0$ for $t \leq s$ and $E(\mathbf{w}_{it}u_{is}) \neq 0$ for $t > s$. Hence, since (a subset of) the regressors might be correlated with the idiosyncratic error term, the standard fixed effect estimator would be inconsistent. To tackle this problem, estimation procedures using instrumental variables (IV) have been extensively used in the empirical literature and (a subset of) lagged level variables, $\mathbf{w}_{i1}, \dots, \mathbf{w}_{it}$, are typically employed as instruments.

In this paper, in order to account for the possible weak exogeneity of the regressors, especially the indicators of access to finance, we rely on the system generalized method of moments (GMM) estimator for panel data developed by Arellano and Bover (1995) and Blundell and Bond (1998), who augment the approach originally proposed by Arellano and Bond (1991).

Indeed, while, under suitable conditions, both the IV and GMM estimators for panel data are consistent and asymptotically normal, the latter was shown to be efficient within the class of instrumental variable estimators, especially when T is small and N large. This occurs for both dynamic and static panel data models because, compared to the IV estimator, the GMM estimator optimally exploits more linear moment restrictions that originate from the assumptions made on the error terms (see e.g. Hayakawa et al., 2019 for a recent contribution on the matter). In our estimation we utilize as instruments the lagged values of the access to finance variables dated $t - 2$ and earlier and, in order to appraise the validity of the instruments, we consider the Hansen test of over-identifying restrictions, which verifies the absence of correlation between the errors and the instruments.

Finally, we cluster the standard errors at the level of provinces, following the advice of Cameron and Miller (2015) to use more aggregate clusters when possible and having checked that our results are confirmed if standard errors are clustered at the industry level.

A parallel specification can be devised for addressing the same issues along the extensive margin of exports. In this case the equation reads as follows:

$$\begin{aligned} DEX_{it} = & \beta_0 + \beta_1 \ln(\text{Exchange rate}_{jt}) + \beta_2 (\text{Access to Finance}_{it-1}) \\ & + \beta_3 \ln(\text{Exchange rate}_{jt}) \times (\text{Access to Finance}_{it-1}) \\ & + \beta_4 \ln(\text{Labor productivity}_{it-1}) + \gamma' Z_{it} + \lambda_i + \psi_t + u_{it}, \end{aligned} \quad (14)$$

where the dependent variable, DEX_{it} , is set equal to one if the firm exports in period t and to zero otherwise. All other variables are as defined as above.

In estimating the previous empirical models, along both the intensive and extensive margins (Eqs. 13 and 14), the existing literature points to a priori, well-established results exclusively as to what pertains the sign of the effects of exchange rate and of access to finance taken in isolation. That is, the estimated coefficient β_1 is expected to be positive in both equations for the standard competitiveness effect: a depreciation increases foreign demand and participation to the export markets. Similarly, the estimated coefficient β_2 in both equations is expected to be positive, as a higher ability to have access to finance, that is a lower incidence of liquidity constraints, is conducive to higher export volumes and a higher probability of being exporter. The intuition is that if two firms are identical except for, say, the ratio of cash holdings to total assets, then the firm with the higher ratio is expected to be better able to meet the working capital requirements associated to exporting activities.

However, in light of the interaction term included in the two equations above, the overall effect of an exchange rate shock on exporting activities depends on the estimated coefficient β_1 plus another term: the estimated parameter β_3 times the value of the indicator of the firm's access to finance. The key prediction of the theoretical model presented in Section 3 is that the standard competitiveness effect of exchange rate on exports is magnified when a firm faces more severe financial constraints.

5. The estimation results

5.1. Results from the baseline specification

5.1.1. The intensive margin

In investigating how exchange rate affects exports differentially across firms depending on their financial constraints, we first focus on the intensive margin. In Table 2 we report the estimation results of the baseline equation in which the log of the volume of exports is the dependent variable. The estimation results are reported for the three alternative measures of financial constraints. The estimated coefficient of the exchange rate in isolation is always positive and statistically significant. This reflects the standard competitiveness effect, that is a currency depreciation drives up the firm's exported volumes. The variable denoting the extent of firm's access to external finance enters both in isolation and interacted with the exchange rate variable. The estimated coefficient associated to the variable in isolation is positive and statistically different from zero in each of the three specifications, suggesting that a firm facing more severe financing constraints (i.e., a lower degree of the "Access to finance" indicator) tends to have lower export volumes, consistent with available empirical evidence in the literature. For example, by looking at the liquidity-to-assets ratio as a proxy of the firm's access to finance (first column), the estimated coefficient is 0.295 with a standard error of 0.107. Importantly, the estimated coefficient of the interaction term is always negative and statistically significant: in the first column, for example, it amounts to -1.552 with a standard error of 0.529. Our empirical findings therefore confirm that, along the intensive margin of exports, a currency depreciation induces a rise of the firm's exports and this effect is stronger the more a firm faces financing constraints.

As for the level of firm's productivity, not surprisingly its estimated effect on exports is positive (and statistically significant) in all the three equations. For example, if the (inverse of the) debt-to-assets ratio is considered (column 3), then the estimated effect is 0.115 with a standard error of 0.028. Regarding the Hansen test of over-identifying restrictions, their corresponding p-values indicate that they are lower than the critical values and this confirms the validity of the empirical specification.

In order to gauge on quantitative grounds how the response of exports to exchange rate differs depending on the firm's degree of financial constraints, we compare the marginal effect of exchange rate on exports evaluated at the firm in the 25th

Table 2

The intensive margin of exports - GMM estimates.

Dependent variable: $\log(\text{Export volumes})_{i,t}$	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{j,t}$	1.902*** (0.358)	3.981*** (0.695)	0.961*** (0.290)
Access to Finance $_{i,t-1}$	0.295*** (0.107)	0.634*** (0.183)	-0.361 (0.797)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to Finance}_{i,t-1}$	-1.552*** (0.529)	-11.068*** (3.187)	-1.652*** (0.587)
$\log(\text{Labor productivity})_{i,t-1}$	0.144*** (0.029)	0.150*** (0.033)	0.115*** (0.028)
Number of observations	688250	521937	861195
Hansen statistic (p-value)	20.7 (0.95)	14.7 (0.62)	16.4 (0.93)
Marginal effects (dy/dx) of :			
$\log(\text{Exchange rate})_{j,t}$ on $\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
evaluated at:			
1) Access to Finance $_{i,t-1}$ (25th percentile)	2.104*** (0.400)	3.175*** (0.511)	1.898*** (0.328)
2) Access to Finance $_{i,t-1}$ (75th percentile)	1.577*** (0.311)	1.089** (0.436)	1.503*** (0.266)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000-2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

percentile of the distribution of the "Access to finance" variable with the marginal effect for the firm in the 75th percentile of the distribution. The results are reported in the bottom panel of Table 2. For example, the firm at the 75th percentile of the distribution of the Assets-to-debt ratio (that is, a firm with relatively low financing constraints) features an estimated effect of exchange rate depreciation on exports of 1.503 (with a standard error of 0.266), a much lower value (by 40 per cent) than that uncovered for the firm at the 25th percentile of the distribution for that variable: 1.898 (with a standard error of 0.328).¹⁴

5.1.2. The extensive margin

In Table 3 we report estimation results for the extensive margin of exports. In particular, the dependent variable is a dummy variable reflecting the firm's export status: it is equal to one if the firm is an exporter in a given year and zero otherwise. The estimation approach for deriving the results reported in Table 3 is that of the linear probability model (LPM) and the GMM panel estimation methodology is employed. The results provide a similar picture to that emerged for the intensive margin. In particular, on the one hand, the standard competitiveness effect is confirmed, suggesting that an exchange rate depreciation increases the probability for a firm to be an exporter. On the other hand, a higher ability for a firm to have access to finance also increases the probability of being exporter. Moreover, our findings show that the firm's participation to export markets is more responsive to exchange rate shocks the more the firm is financially constrained. For example, by focusing on the net liquidity-to-total assets ratio as proxy for access to finance, the estimated coefficient of the exchange rate variable in isolation is 0.197 (with a standard error of 0.033). The coefficient on the interaction term between the exchange rate and the access to finance indicator is negative and statistically significant (-0.102 with a standard error of 0.043), suggesting that if the firm's access to finance is relatively high, then the sensitivity of export participation to exchange rate is attenuated. We also report the marginal effects of the exchange rate on the probability of exporting for two different firms: the one at the 25th percentile of the distribution of the access to finance variable and the one at the 75th percentile. For all indicators of the ability to have access to finance, we find that the more a firm is liquidity constrained, the stronger is the marginal effect.

To investigate the sensitivity of export participation to exchange rate depending on the degree of firms' financial constraints, we also experimented with the probit estimation approach with instrumental variables (IV). From a qualitative point of view, the empirical results obtained with this approach are similar to those obtained with the LPM model. As documented in Table 4, we find that no matter which indicator of access to finance we use, the marginal effect of exchange on the probability of exporting is higher for the firms with a lower ability to have access to external finance. For example, when

¹⁴ As we discuss in the Introduction, the difference between our results and those of Ma et al. (2019) and Li et al. (2020) is not surprising, since our empirical model focuses on aggregate firm exports and not bilateral exports towards one specific country. The estimated exchange rate elasticities of exports are slightly larger than those reported by Berman et al. (2012) in their study on the effect of productivity on the exchange rate sensitivity of French firms' exports.

Table 3

The extensive margin of exports: Linear Probability Model – GMM estimates.

Dependent variable: Export status _{it}	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity</i> <i>Assets</i>	<i>Cash</i> <i>Assets</i>	<i>Assets</i> <i>Debt</i>
log(Exchange rate) _{jt}	0.197*** (0.033)	0.271*** (0.037)	0.039*** (0.017)
Access to Finance _{it-1}	0.030*** (0.008)	0.032** (0.013)	0.085*** (0.017)
log(Exch. rate) _{jt} · Access to Finance _{it-1}	−0.102** (0.043)	−0.583*** (0.067)	−0.083** (0.042)
log(Labor productivity) _{it-1}	0.012*** (0.003)	0.086*** (0.017)	0.001 (0.001)
Number of observations	591268	885245	886195
Hansen statistic (p-value)	13.6 (0.99)	14.7 (0.95)	15.6 (0.62)
Marginal effects (dy/dx) of : log(Exchange rate) _{jt} on prob(Exporting) _{it} evaluated at:	Measures of Financial constraints		
	(1)	(2)	(3)
1) Access to Finance _{it-1} (25th percentile)	0.209*** (0.037)	0.231*** (0.034)	0.086*** (0.0157)
2) Access to Finance _{it-1} (75th percentile)	0.175*** (0.027)	0.122*** (0.027)	0.066*** (0.011)

Notes: The dependent variable is the probability of being exporter. It is a dummy variable equal to one if, in a given year, the firm has exported and zero otherwise. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm's ability to have access to external finance. Linear Probability Models are estimated here using the GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

the Assets-to-debt ratio is considered, the estimated marginal effect is 0.225 (with a standard error of 0.006) for the firm at the 25th percentile of the distribution for that variable, while it is 0.156 (with a standard error of 0.007) for the firm at the 75th percentile (the one with lower financing constraints).

To sum up, by using our baseline empirical framework and using a variety of measures for the firms' financial conditions, we provide the following findings. First, we confirm that financial constraints hamper both the volumes of exports and the decision to participate to exporting activities. Second, we show that a depreciation, by inducing a higher competitiveness, does increase the export volumes and rise the probability of exporting. Moreover, we establish that for a firm facing financial restraints the expansionary effect of an exchange rate depreciation on exporting activities is magnified and this holds true at both the intensive and extensive margin of exports and for all our alternative measures of access to finance. Building on these findings, in the rest of the paper we do the following. First, we try to shed light on the mechanisms through which these effects arise. Second, we investigate whether the incidence of financial constraints continues to shape the elasticity of firms' exports to exchange rate when other transmission channels are explicitly allowed for.

5.2. Deciphering the mechanism: the exchange rate pass-through

As argued above and as emerged in our simple theoretical framework, a key mechanism through which financial constraints affect the responsiveness of firm's exporting activities to exchange rate is the degree of exchange rate pass-through to export prices. As argued by [Strasser \(2013\)](#), more financially constrained firms might tend to pass-through exchange rate shocks to export prices denominated in the foreign currency by more than what the less constrained firms do, causing stronger swings in export volumes. This is because borrowing constraints force firms to keep pricing-to-market decisions to a minimum as they have not enough margin to absorb an exchange rate shock on their mark-up. Hence, the exchange rate pass-through to export prices denominated in the domestic currency is low for these firms and export volumes respond to a larger extent. Our data allow us to investigate the empirical relevance of this transmission channel of an exchange rate shock. In particular, we can rely on firm-level, time-varying information on the unit value of exports of Chinese firms denominated in the domestic currency.¹⁵

For this pass-through analysis we employ the logarithm of firm-level, time-varying unit value of exports of Chinese firms denominated in the domestic currency as the dependent variable in a regression equation otherwise identical to Eqs. (13) and (14).

¹⁵ See [Silver \(2009\)](#) for a critical assessment on the use of unit values of imports and exports as proxies for import and export prices.

Table 4

The extensive margin of exports - Probit Estimates with IV.

Dependent variable: Export status _{i,t}	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
log(Exchange rate) _{j,t}	0.500*** (0.018)	0.374*** (0.024)	0.407*** (0.019)
Access to Finance _{i,t-1}	0.264*** (0.011)	5.532*** (0.131)	0.167*** (0.058)
log(Exch. rate) _{j,t} · Access to Finance _{i,t-1}	−0.534** (0.048)	−0.920*** (0.106)	−1.460*** (0.119)
log(Labor productivity) _{i,t-1}	0.064*** (0.002)	0.006* (0.003)	0.059*** (0.002)
Number of observations	537608	563695	504524
Marginal effects (dy/dx) of : log(Exchange rate) _{j,t} on prob(Exporting) _{i,t} evaluated at:			
1) Access to Finance _{i,t-1} (25th percentile)	0.128*** (0.004)	0.056*** (0.004)	0.225*** (0.006)
2) Access to Finance _{i,t-1} (75th percentile)	0.092*** (0.005)	0.048*** (0.008)	0.156*** (0.007)

Notes: The dependent variable is the probability of being exporter. It is a dummy variable equal to one if, in a given year, the firm has exported and zero otherwise. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Industry dummy variables are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. The instrument set includes lagged values of the financial indicator. Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

The estimation results are reported in Table 5 and confirm the empirical relevance of the channel based on differences across firms in the degree of exchange rate pass-through to export prices. In particular, we first find that an exchange rate depreciation increases the export prices denominated in the firm's domestic currency (the renminbi). Moreover, and importantly for our purposes, we establish that the higher is the firm's access to finance the stronger is the effect of a currency depreciation on the export price denominated in the domestic currency and, therefore, the lower is (the absolute value of) the effect on export prices denominated in the foreign currency. In particular, we find that the estimated coefficient on the interaction term between the log of the exchange rate and the indicator of access to finance is positive and statistically significant. For example, if the Cash-to-total assets ratio is used (column 2), then the marginal effect of (the log of the) exchange rate on the log of Export prices in renminbi is 0.84 (with a standard error of 0.295) for the firm at the 25th percentile of the distribution of the access to finance variable. By contrast, the marginal effect is equal to 1.118 (with a standard error of 0.237) for the firm at the 75th percentile of the distribution. These regressions include the lagged log-level of labor productivity and, interestingly enough, its estimated effect on the export prices denominated in the domestic currency is in general negative and statistically significant.¹⁶

6. Allowing for alternative transmission channels

A critic might argue that our finding on the importance of access to finance in shaping the exchange rate sensitivity of exports is a figment of other factors. Indeed, several firm-level, time-varying features, in addition to financial constraints, may affect the responsiveness of exports to exchange rate and controlling for them is crucial to strengthen our causal interpretation based on financial constraints. Hence, in this section we investigate whether our results continue to hold once we augment the baseline specification to allow for alternative transmission mechanisms. In particular, we consider, each at a time, the role of imported intermediate input, labor productivity, price stickiness, and size.

6.1. Import dependence

The first mechanism that we consider deals with the degree of firm's reliance on imported intermediate inputs. While an exchange rate depreciation makes the domestic products more competitive abroad as their foreign export prices become cheaper, it also makes the firm's imported inputs more expensive. This offsetting effect of exchange rate shocks through imported inputs is likely to be more pronounced the higher is the firm's dependence on imported intermediate inputs and may have implications for the overall effect on exports. Greenaway et al. (2010) provide evidence that the negative

¹⁶ The choice of the invoicing currency in a trade transaction affects the degree of exchange rate pass-through to export prices and the responsiveness of exports (see e.g. [Gopinath, 2016](#)). Moreover, a stream of literature has established that this choice is endogenous and a number of factors can contribute to shape it (see e.g. [Goldberg and Tille, 2016](#)). Whilst the issue is relevant, we do not address it in the paper.

Table 5
Inspecting the mechanism: the pass-through channel (GMM).

Dependent variable: $\log(\text{Export prices in domestic currency})_{j,t}$	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{j,t}$	1.190*** (0.271)	0.712** (0.345)	1.417*** (0.336)
Access to Finance $_{i,t-1}$	-0.273* (0.156)	-0.344 (0.806)	0.101*** (0.029)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to Finance}_{i,t-1}$	0.618** (0.293)	1.499* (0.886)	0.948*** (0.364)
$\log(\text{Labor productivity})_{i,t-1}$	0.223 (0.258)	-0.058*** (0.015)	-0.052*** (0.013)
Number of observations	133191	104164	119026
Hansen statistic (p-value)	12.1 (0.28)	13.7 (0.84)	11.42 (0.78)
Marginal effects (dy/dx) of :			
$\log(\text{Exchange rate})_{j,t}$ on $\log(\text{Export prices})_{j,t}$			
evaluated at:			
1) Access to Finance $_{i,t-1}$ (25th percentile)	1.144*** (0.263)	0.840*** (0.295)	0.906*** (0.209)
2) Access to Finance $_{i,t-1}$ (75th percentile)	1.335*** (0.302)	1.118*** (0.237)	1.120*** (0.252)

Notes: The dependent variable is the (log) of firm's export prices denominated in domestic currency. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t - 1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

response on exports of an exchange rate appreciation is smaller in industries that import a higher share of intermediate inputs because of this offsetting effect on the cost side through the price of inputs. In a recent contribution, [Amiti et al. \(2014\)](#) show that the degree of exchange rate pass-through to export prices depends on the firm's import intensity and export market share. In particular, a key prediction of their theoretical model, which is also confirmed at the empirical level using Belgian microeconomic data, is that the exchange rate pass-through into destination-currency export prices is lower for more import-intensive firms (and for firms with higher export share in the destination market). Whilst [Amiti et al. \(2014\)](#) do not explicitly investigate the implications for export volumes of differences across firms in their reliance on imported inputs, their findings single out a channel through which the sensitivity of exports to exchange rate differs across firms. In particular, firms' heterogeneity in the intensity of imported intermediates does play a crucial role in their framework.

We construct a firm-level, time varying indicator of firm's net import dependence. This indicator is constructed as the share of imported intermediate inputs (net of exports) over total sales. In [Table 6](#) we provide the results of estimating an equation where, in addition to the channel based on the incidence of financing restraints, we also allow for the firm's net import dependence. This variable also enters the specification both in isolation and interacted with the (log of the) exchange rate. To mitigate the possible endogeneity bias, the lag of net import dependence is inserted in the specification. Consistently with the results of [Amiti et al. \(2014\)](#) and [Greenaway et al. \(2010\)](#), we find that a higher degree of firm's reliance on imported inputs lowers the sensitivity of exports to exchange rate: indeed, in all the three specifications documented in [Table 6](#) the estimated coefficient on the interaction term is negative and statistically significant. Importantly for our purposes, however, a differential effect of exchange rate on exports channeled through the ability of firms to have access to finance continues to be detected. The estimated coefficients on the interaction term between the indicator of access to finance and the exchange rate is negative and statistically significant no matter which indicator is used: for example, if the assets to debt ratio is considered, then the estimated coefficient on the interaction term is -2.097 (with a standard error of 1.068). The marginal effects of the log of exchange rate on the log of export volumes are indeed different across firms depending on their position within the distribution of the access to finance variables (see the figures on [Table 6](#) for a comparison between firms at the 25th and 75th percentile of the distribution).

As in all previous equations presented thus far, the log of labor productivity is included in the specifications whose results are reported in [Table 6](#) and, not surprisingly, the estimated effect of it continues to be positive and statistically significant. In the next section we delve deeper into the role of productivity.

6.2. Firm's productivity

The second alternative mechanism that we consider deals with the role of productivity. As mentioned above, in investigating the export decision at the firm level a prominent role is assigned in the literature to the observed cross-sectional heterogeneity in productivity. The latter originates from several sources, including for example: (a) a continuous develop-

Table 6

Alternative Explanations: Import Dependence (GMM estimates).

Dependent variable: log(Export volumes) _{it}	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
log(Exchange rate) _{jt}	1.942*** (0.367)	3.242*** (0.586)	1.017*** (0.344)
Access to Finance _{it-1}	0.298*** (0.107)	0.607*** (0.184)	1.122* (0.632)
log(Exch. rate) _{jt} · Access to Finance _{it-1}	-1.551*** (0.534)	-8.404*** (2.714)	-2.097** (1.068)
Net Import Dependence _{it-1}	-0.0003* (0.0001)	-0.0003** (0.0002)	-0.0003** (0.0001)
log(Exch. rate) _{jt} · Net Import Dependence _{it-1}	-0.002** (0.001)	-0.002** (0.001)	-0.002* (0.001)
log(Labor productivity) _{it-1}	0.127*** (0.028)	0.111*** (0.028)	0.103*** (0.032)
Number of observations	680068	784226	638266
Hansen statistic (p-value)	11.1 (0.99)	13.9 (0.84)	14.9 (0.72)
Marginal effects (dy/dx) of : log(Exchange rate) _{jt} on log(Export volumes) _{it} evaluated at:	Measures of Financial constraints		
	(1)	(2)	(3)
1) Access to Finance _{it-1} (25th percentile)	2.146*** (0.410)	2.657*** (0.436)	2.203*** (0.436)
2) Access to Finance _{it-1} (75th percentile)	1.617*** (0.319)	1.069*** (0.371)	1.719*** (0.270)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Net Import Dependence is defined in the text. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t - 1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

ment of new products and new production techniques, which generate very differentiated outcomes in terms of efficiency gains; (b) differences in entrepreneurial and managerial ability; (c) an uneven diffusion of information and knowledge, which may favor some firms over others. More productive firms are shown to be better able to pay the fixed, start-up costs associated with export participation and, in general, higher levels of firm's productivity are associated with larger export flows (see Melitz (2003) and Bernard et al. (2007) for a survey).

In their important contribution, Berman et al. (2012) show that firms with higher productivity exhibit a lower elasticity of exchange rate pass-through into export prices (denominated in the currency of the destination country) because they tend to absorb more exchange rate movements in their mark-ups and this implies a lower responsiveness of their export volumes to exchange rate. Hence, in their analysis differences across firms in the sensitivity of exports to exchange rate originate from the heterogeneity across firms in their productivity (see Li et al. (2015) for a similar analysis on Chinese data). To allow for this additional channel, we therefore augment our baseline specification to include the interaction between the log level of the exchange rate and the log level of lagged labor productivity. As before, labor productivity in isolation is also inserted in the estimating equations.

The results are reported in Table 7. We confirm the main finding of Berman et al. (2012), as the interaction between exchange rate and productivity is negative throughout the three specifications (although it is not statistically significant when the liquid-to-total assets ratio is considered). For example, when the assets-to-debt ratio is used, the estimated coefficient on the interaction term is -0.211 (with a standard error of 0.072). However, the mechanism through which financial constraints affects the response of export to exchange rate continues to be relevant in our estimates. Indeed, the estimated coefficient associated to the interaction term between the exchange rate and the indicator of access to finance is negative and statistically significant across all indicators of (lack of) financial restraints. By focusing on the marginal effects of exchange rate on exports for firms at the 25th and 75th percentile in the distribution of, say, the liquidity-to-assets ratio, their estimates are, respectively, 2.101 (with a standard error of 0.396) and 1.616 (with a standard error of 0.321). Our results indicate that, even allowing for the channel based on productivity, the higher are the financial constraints the larger the sensitivity of exports to exchange rate tends to be.¹⁷

¹⁷ Financial constraints and productivity have also been shown to interact with each other in determining export behavior. Berman and Hericourt (2010), for example, show that the level of productivity becomes increasingly relevant for exporting activity when financial constraints are relatively low, while it does not affect export decisions when access to finance is limited. They argue that, as the correlation between productivity and access to finance might be imperfect, a disconnection can arise between exports on one side and productivity and access to finance on the other if the latter two variables are not jointly considered.

Table 7
Alternative Explanations: Productivity Performance (GMM estimates).

Dependent variable: $\log(\text{Export volumes})_{i,t}$	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{j,t}$	2.467*** (0.599)	4.639*** (0.950)	1.761*** (0.433)
Access to Finance $_{i,t-1}$	0.295*** (0.107)	0.630*** (0.181)	0.721* (0.434)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to Finance}_{i,t-1}$	-1.428** (0.566)	-10.910*** (3.172)	-1.609** (0.779)
$\log(\text{Labor productivity})_{i,t-1}$	0.142*** (0.030)	0.153*** (0.032)	0.103*** (0.029)
$\log(\text{Exch. rate})_{j,t} \cdot \log(\text{Labor productivity})_{i,t-1}$	-0.151 (0.105)	-0.181* (0.097)	-0.211*** (0.072)
Number of observations	688250	521937	886195
Hansen statistic (p-value)	19.7 (0.97)	13.1 (0.73)	17.1 (0.58)
Marginal effects (dy/dx) of :			
$\log(\text{Exchange rate})_{j,t}$ on $\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
evaluated at:			
1) Access to Finance $_{i,t-1}$ (25th percentile)	2.101*** (0.396)	3.159*** (0.507)	1.896*** (0.351)
2) Access to Finance $_{i,t-1}$ (75th percentile)	1.616*** (0.321)	1.102** (0.436)	1.511*** (0.222)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000-2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

6.3. Price stickiness

An additional channel that may contribute to shape the elasticity of exports to exchange rate changes deals with nominal rigidities. As argued by [Gopinath and Itskhoki \(2010\)](#), if the prices in the local currency of the destination market are sticky then the sensitivity of export volumes to exchange rate tends to be low. Indeed, if in the short run firms do not adjust the local currency export prices, then the exchange rate pass-through on export prices denominated in the domestic currency is high and limited would be the response of export volumes (see also [Gopinath and Rigobon \(2008\)](#)).

In order to allow for this additional channel, we follow [Berman et al. \(2012\)](#) and use the standard deviation of the absolute value of log changes in the unit value of exports (in the home currency) as an admittedly imperfect proxy for the frequency of price revisions. In particular, we insert in the baseline specification two additional terms: a) the standard deviation in isolation and b) its interaction with the exchange rate. The results are documented in [Table 8](#). As expected, a higher variability over time in the log change of the unit value of exports denominated in the domestic currency – which reflects a lower frequency of adjustment in the buyer currency prices – implies a lower effect of exchange rate on exports. For example, in the equation where the liquid-to-total assets ratio is used as indicator of access to finance, the interaction term between the exchange rate and the proxy for price rigidity has a coefficient which is estimated to be negative and statistically significant (-2.153 with a standard error of 0.519). Against this backdrop, however, our main results remain unchanged, as the sensitivity of exports to exchange rate continues to vary depending on the firm's ability to access external finance.

6.4. Employment size

Finally, the last channel that we investigate deals with firm size. As mentioned before, our empirical specifications include different dummy variables to control for the firm's employment size. In this section we verify whether financial constraints continue to play an independent and statistically significant role in shaping the exchange rate elasticity of exports once the number of employees is introduced in the regression both in isolation and interacted with the exchange rate. Controlling for size also is necessary in order to rule out the possibility that the influence of financial constraints is simply a figment of other firm-level factors. In particular, firm size can be thought as being isomorphic to other firm level features, such as financial

Table 8
Alternative Explanations: Price Rigidity (GMM Estimates).

Dependent variable: $\log(\text{Export volumes})_{i,t}$	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{j,t}$	2.765*** (0.448)	4.163*** (0.547)	1.840*** (0.695)
Access to Finance $_{i,t-1}$	0.430*** (0.114)	1.113** (0.457)	-2.439** (0.979)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to Finance}_{i,t-1}$	-2.978*** (0.946)	-6.983** (3.249)	-2.574** (1.069)
$\log(\text{Exch. rate})_{j,t} \cdot \text{st.dev.} \Delta \text{Export_prices}_i $	-2.153*** (0.519)	-2.500*** (0.542)	-2.256*** (0.451)
$\log(\text{Labor productivity})_{i,t-1}$	0.178*** (0.046)	0.233*** (0.038)	0.629*** (0.131)
Number of observations	147389	118718	178620
Hansen statistic (p-value)	11.23 (0.99)	12.9 (0.74)	13.6 (0.99)
Marginal effects (dy/dx) of :			
$\log(\text{Exchange rate})_{j,t}$ on $\log(\text{Export volumes})_{i,t}$ evaluated at:	(1)	(2)	(3)
1) Access to Finance $_{i,t-1}$ (25th percentile)	1.884*** (0.456)	2.318*** (0.363)	2.099*** (0.342)
2) Access to Finance $_{i,t-1}$ (75th percentile)	0.961*** (0.272)	1.016** (0.463)	1.514*** (0.359)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. The measure of price rigidity, the standard deviation of $|\Delta \text{Export_prices}_i|$, is discussed in the text. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

strength, and is also often used in empirical analyses to approximate the firm's propensity to hedge against exchange rate risk.¹⁸

In the context of our investigation, the higher is the reliance on foreign exchange hedging activities the lower would be the responsiveness of exports to exchange rate shocks. Since we do not have comprehensive hedging data at the firm level, we use data on employment size and extend our baseline specification accordingly. The results are reported in Table 9. As expected, the estimated coefficient on the interaction term between (the log of the) exchange rate and the log of lagged employment is negative and statistically significant no matter which indicator of access to finance is used. For example, when the cash-to-assets ratio is used, the estimated coefficient of the interaction term is -5.724 (with a standard error of 1.390). Importantly, however, exchange rate continues to affect export volumes differentially across firms depending on their degree of access to finance. For example, if a firm has a liquid-to-total assets ratio at the 25th percentile of the distribution then the marginal effect of exchange rate on exports is 2.432 (with a standard error of 0.686). Conversely, for the firm at the 75th percentile of that distribution (a firm with lower financial constraints) the marginal effect is 1.369 (with a standard error of 0.788).

Hence, these results confirm that, overall, several factors different from financial constraints contribute to explain the heterogeneity across firms in the response of exports to exchange rate. However, we have shown that, even controlling for each of them, differences across firms in the ability to access external finance introduce a statistically significant degree of specificity in the response of their exports to exchange rate.

7. Extensions and robustness checks

7.1. Depreciation vs appreciation of the currency

The analysis presented thus far does assumes that the effect of exchange rate variations on exports operate symmetrically, without allowing for the response of the firms' exports to differ in the cases of depreciations and appreciations. Against this backdrop, it is therefore useful to verify whether this assumption survives a more careful data scrutiny.

¹⁸ Hericourt and Poncet (2013) document a non-linear effect of exchange rate volatility on firm-level export performance depending on the extent of financial constraints. They show that, because well-developed financial markets allow firms to hedge exchange rate risk, the negative impact of exchange rate volatility on exports is amplified for financially vulnerable firms and attenuated if the level of financial development is high.

Table 9
Alternative Explanations: Employment Size (GMM estimates).

Dependent variable: log(Export volumes) _{i,t}	Measures of Financial constraints		
	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
log(Exchange rate) _{j,t}	49.669*** (13.504)	30.692*** (7.230)	14.531*** (3.447)
Access to Finance _{i,t-1}	0.360*** (0.106)	0.681*** (0.166)	0.754** (0.381)
log(Exch. rate) _{j,t} · Access to Finance _{i,t-1}	−3.082*** (0.747)	−6.398*** (1.780)	−3.948*** (0.960)
log(Employment size) _{i,t-1}	0.925*** (0.199)	0.805*** (0.146)	0.871*** (0.137)
log(Exch. rate) _{j,t} · log(Employment size) _{i,t-1}	−9.515*** (2.734)	−5.724*** (1.390)	−3.021*** (0.730)
Number of observations	716338	930391	926055
Hansen statistic (p-value)	17.6 (0.06)	21.9 (0.64)	17.1 (0.11)
Marginal effects (dy/dx) of : log(Exchange rate) _{j,t} on log(Export volumes) _{i,t} evaluated at:	Measures of Financial constraints		
	(1)	(2)	(3)
1) Access to Finance _{i,t-1} (25th percentile)	2.432*** (0.686)	2.209 (0.502)	2.047*** (0.418)
2) Access to Finance _{i,t-1} (75th percentile)	1.369* (0.788)	1.010** (0.412)	1.103*** (0.273)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t - 1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

To this purpose, we modify our baseline specifications and estimate distinctly the exchange rate sensitivity of exports in the case of a currency depreciation and of an appreciation. We do so by constructing two dummy variables: D_{dep} , which is equal to one if the industry exchange rate has depreciated in a given year (and zero otherwise) and D_{app} , equal to one if an appreciation has occurred in a given year (and zero otherwise). Each of these dummy variables are interacted with the (log) level of the industry exchange rate taken in isolation and with the interaction between the exchange rate itself and the lagged indicator of access to finance.

The results, presented in Table 10, clearly indicate that the lower is the firm's ability to have access to finance the stronger are both the positive response of exports to an exchange rate depreciation and the negative response of them to an appreciation. The estimated coefficients associated with the exchange rate level is positive and statistically significant, no matter whether the industry-specific currency value depreciated or appreciated in a given year. Similarly, the estimated coefficient associated with the interaction term between the exchange rate and the indicator of access to finance is negative and statistically significant in both cases. For example, when the financial indicator based on liquidity is used, the estimated coefficient of the interaction term is -0.747 (with a standard error of 0.348) if an appreciation occurs and it is -0.903 (with a standard error of 0.284) in the case of a depreciation.

The results on the marginal effects at different points of the distribution across firms of the access to finance variable also support the hypothesis of no asymmetrical effects of depreciations and appreciations. Irrespective of whether we consider depreciations or appreciations, the estimated response of exports to an exchange rate variation is larger (in absolute value) if the firm's ability to access finance is relatively low (the value of the indicator at the 25th percentile) compared to the case in which it is relatively high (the value at the 75th percentile). This finding holds true for all measures of firm's financial condition. For example, when we consider the assets-to-debt ratio, in the case of depreciations the estimated marginal effect on exports of an exchange rate variation is 1.640 (with a standard error of 0.292) for the firm at the 25th percentile of the distribution for that variable, while it is 1.189 (with a standard error of 0.192) for the firm at the 75th percentile (the one with lower financing constraints). Similarly, in the case of appreciations also the marginal effect of the exchange rate variation is stronger if the firm faces financing constraints. In particular, the estimated effect is 2.349 (with a standard error of 0.453) for the firm at the 25th percentile of the distribution, while it is 1.878 (with a standard error of 0.288) for the firm at the 75th percentile (see Table 10).

Overall, these findings reinforce our conclusion that the presence of credit rationing in a firm amplifies, in a statistically significant way, the size of the positive and negative response of its exports to, respectively, a currency depreciation and appreciation. This is reassuring in that, differently from our own case study, in a given sample period there might be a

Table 10
Allowing for Differential Effects of Exchange Rate Depreciations and Appreciations.

Dependent variable: $\log(\text{Export volumes})_{i,t}$	Measures of Financial constraints		
	(1) <i>Liquidity</i> <i>Assets</i>	(2) <i>Cash</i> <i>Assets</i>	(3) <i>Assets</i> <i>Debt</i>
$\log(\text{Exchange rate})_{j,t} \cdot D_{dep}$	1.682*** (0.452)	2.841*** (0.588)	0.570** (0.242)
$\log(\text{Exchange rate})_{j,t} \cdot D_{app}$	1.090*** (0.271)	7.071*** (1.670)	1.232*** (0.220)
$\text{Access to finance}_{i,t-1}$	0.277*** (0.102)	0.729*** (0.177)	0.592 (0.387)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to finance}_{i,t-1} \cdot D_{dep}$	-0.903*** (0.284)	-6.860*** (2.233)	-1.886*** (0.687)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to finance}_{i,t-1} \cdot D_{app}$	-0.747*** (0.412)	-24.983*** (9.426)	-1.969** (0.854)
$\log(\text{Labor productivity})_{i,t-1}$	-0.078 (0.074)	0.149*** (0.033)	0.105*** (0.029)
Number of observations	925916	521937	886195
Hansen statistic (p-value)	13.7 (0.99)	11.9 (0.37)	10.4 (0.98)
Marginal effects (dy/dx) of : $\log(\text{Exchange rate})_{j,t} \cdot D_{dep}$ on $\log(\text{Export volumes})_{i,t}$ evaluated at:			
	(1)	(2)	(3)
1) Access to finance _{i,t} (25th percentile)	1.205*** (0.298)	2.342*** (0.459)	1.640*** (0.292)
2) Access to finance _{i,t} (75th percentile)	0.898*** (0.231)	1.049*** (0.329)	1.189*** (0.192)
Marginal effects (dy/dx) of : $\log(\text{Exchange rate})_{j,t} \cdot D_{app}$ on $\log(\text{Export volumes})_{i,t}$ evaluated at:			
	(1)	(2)	(3)
1) Access to finance _{i,t} (25th percentile)	1.778*** (0.486)	5.252*** (1.023)	2.349*** (0.453)
2) Access to finance _{i,t} (75th percentile)	1.523*** (0.402)	0.543 (0.955)	1.878*** (0.288)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. D_{dep} and D_{app} are dummy variables equal to one (and zero otherwise) if, in year t , the industry exchange rate, respectively, depreciates and appreciates. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using an instrumental variable panel data estimator different from the GMM's. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set includes lagged values of it. Sample period: 2000–2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

predominance of episodes of currency appreciations or of depreciations. If according to the empirical evidence the response of exports to exchange rate were found to be asymmetric then a critic might argue that the estimated response of exports to exchange rate is a figment of that predominance in the sample. Our findings therefore suggest that this contention would not be compelling.

7.2. Firm-level exchange rates

In analyzing thus far how the responsiveness of export participation to exchange rate is shaped by the firm's ability to access finance, we have relied on industry-specific exchange rates, constructed with trade weights at the industry level. In their paper on how exchange rate shocks affect employment, Dai and Xu (2017) use firm-specific exchange rates to allow for heterogeneity across firms in their export destinations and import source countries. The idea is intuitive: as firms differ in their trading patterns, the effective exchange rate for each of them are also different, and thus firm-level exchange rate shocks should be considered.

Against this backdrop, we have opted for industry-specific exchange rates in the analysis of previous sections for two main reasons. First, an important dimension of export participation in our investigation is the firm's decision of being exporter, i.e. the extensive margin. If this dimension is considered, it is impossible to employ firm-level exchange rates.¹⁹ In fact, because firm-level exchange rates are available only for firms engaged in international trade, their use implies that the sample size available for the analysis is significantly smaller. Second, as firm's exports are our dependent variable and are also used in

¹⁹ Clearly, this is different from the reason to enter a specific foreign market, which is the extensive margin studied, for example, by Ma et al. (2019) and Li et al. (2020).

the trade weights for computing firm-specific exchange rates, if the latter are employed then some concerns on endogeneity may arise.

Despite these limitations, we have verified the robustness of our results by re-estimating the baseline empirical specifications, for each single measure of financial constraints, using firm-level exchange rates. Reassuringly, the main findings of the paper continue to hold also when firm-specific exchange rate movements are considered. The estimates reported in Table 11 clearly confirm that, no matter which indicator of access to finance is used, an exchange rate depreciation (appreciation) causes export volumes to rise (drop) and the effect is stronger the more a firm faces financial constraints. For example, when the liquidity-to-assets ratio is used, the estimated coefficient of the firm-level exchange rate level in isolation is positive and statistically significant, while that of the interaction between the exchange rate and the indicator of liquidity is negative and significant (0.813 with a standard error of 0.293). If the firm exhibits a relatively low value of the liquidity-to-assets ratio (25th percentile), the marginal effect of a firm-specific exchange rate shock on exports is estimated to be higher than that detected for a firm with a relatively high value of it (75th percentile). In the first case the marginal effect is 1.365 (with a standard error of 0.492), while in the second it is 1.113 (with a standard error of 0.506).

In addition to estimating the baseline specifications on the subsample of firms which conduct international trade in a given year, and for which it is therefore possible to calculate the real effective exchange rate, we have also experimented an alternative empirical strategy, using firm-level exchange rates when available and imputing the industry-specific exchange rate for all other firms. The results, available upon request, are qualitatively very similar to those of the baseline specifications.²⁰

7.3. Processing trade

In Section 6.1 we have shown that – for a given extent of access to finance – the positive response of exports to an exchange rate depreciation tends to be mitigated the more a firm is reliant on imported intermediate inputs. Those results lend support to the view that, although access to finance is shown to induce a heterogeneous response of exports to exchange rate, also the external orientation of the firm does matter in shaping the effect of exchange rate on exports. In that specification, the degree of firm's international exposure on the import side was approximated by net import dependence, i.e. the share of intermediate inputs (net of exports) to total sales. It is, however, important to control also for another dimension of external orientation, which relates to the involvement of a firm in the global value chains, when different stages of the production process are dispersed in multiple countries. A specific configuration of this production fragmentation is processing trade, which can be defined as the activity of assembling imported intermediate inputs into final goods and exporting them. Since processing trade is widespread in modern economies, and especially in China (see e.g. Dai et al., 2016), we also investigate if the main results documented earlier continue to hold when we explicitly control whether a firm is engaged in processing trade transactions and the extent to which this occurs. To tackle this issue, we construct a dummy variable that is equal to one if a firm is recorded in the data collected and maintained by the China's General Administration of Customs as engaging only in processing transactions in a given year and zero otherwise (we refer to it as a “processing firm”). In Table 12 we document that the main findings of the paper are broadly unchanged when we include a dummy variable, whose estimated coefficient is positive and statistically significant, for firms that in the previous year only engaged in processing transactions. In particular, as both the estimated coefficients and marginal effects show in the table, a positive exchange rate swing (i.e. a depreciation) rises exports and the impact of it is more pronounced the weaker are the firm's financial conditions. This holds true no matter which indicator of access to finance is used in the corresponding regression specification.

As a further robustness test of our main findings, we also experimented with several alternative ways of controlling for the firm's processing status. First, we used a different dummy variable to identify processing firms, which is equal to one if a firm is engaged in both processing and non-processing transactions and zero otherwise. Second, we focused on an “intensive margin” measure of processing trade, namely a continuous variable given by the share of firm's processing exports to its total exports, also available from data provided by China's General Administration of Customs. Third, we constructed a firm-level indicator of external orientation that reflects the incidence of processing trade of a firm: $\frac{|Imports_{it} - Exports_{it}|}{Imports_{it} + Exports_{it}}$. The indicator lies between zero and one and it takes lower values for firms that have higher values of both import and export transactions, as it happens in the case of processing trade. Overall, irrespective of which of these alternative ways to characterize processing trade is used, the estimation results indicate that our main findings of the paper are generally confirmed.²¹

8. Concluding remarks

In this paper, we investigate on microeconomic data how the reaction of exporting activities to exchange rate movements varies across firms depending on the degree of financial constraints that they face. Consistent with the testable implications of our simple theoretical framework, we find that exporting activities by firms with more financial restraints tend to be more

²⁰ For example, if the assets-to-debt ratio is considered (i.e. the inverse of leverage), then the estimated marginal effect of exchange rate on exports for the firm at the 25th percentile of this indicator (with relatively high financial constraints) is positive and statistically significant (1.011 with a standard error of 0.346), while for the firm at the 75th percentile, the marginal effect, whilst positive, is not statistically significant (0.233 with a standard error of 0.481).

²¹ These additional findings are not reported here due to space constraints but they can be made available upon request.

Table 11
Using Firm-Specific Exchange Rates - GMM Estimates.

Dependent variable:	Measures of Financial constraints		
$\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{i,t}$	1.321*** (0.493)	0.705** (0.308)	-0.121 (0.990)
Access to finance $_{i,t-1}$	0.400*** (0.144)	0.813*** (0.267)	4.260*** (1.100)
$\log(\text{Exch. rate})_{i,t} \cdot \text{Access to finance}_{i,t-1}$	-0.813*** (0.293)	-0.332 (0.438)	-2.362* (1.382)
$\log(\text{Labor productivity})_{i,t-1}$	0.446*** (0.036)	0.399*** (0.037)	0.877*** (0.065)
Number of observations	69021	106536	63399
Hansen statistic (p-value)	16.4 (0.23)	12.7 (0.55)	9.6 (0.30)
Marginal effects (dy/dx) of :	Measures of Financial constraints		
$\log(\text{Exchange rate})_{i,t}$ on $\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
evaluated at:			
1) Access to finance $_{i,t}$ (25th percentile)	1.365*** (0.492)	0.677** (0.299)	1.131*** (0.336)
2) Access to finance $_{i,t}$ (75th percentile)	1.113** (0.506)	0.613** (0.295)	0.578 (0.603)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the firm level. In each year, the sample refers to exporting firms only. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000-2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

Table 12
Controlling for the firm's processing status - GMM Estimates.

Dependent variable:	Measures of Financial constraints		
$\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
	<i>Liquidity Assets</i>	<i>Cash Assets</i>	<i>Assets Debt</i>
$\log(\text{Exchange rate})_{j,t}$	1.564*** (0.267)	3.579*** (0.454)	0.870* (0.509)
Access to finance $_{i,t-1}$	0.243*** (0.074)	0.561*** (0.196)	-1.561 (1.427)
$\log(\text{Exch. rate})_{j,t} \cdot \text{Access to finance}_{i,t-1}$	-0.650* (0.359)	-9.772** (2.655)	-2.325 (1.606)
$\log(\text{Labor productivity})_{i,t-1}$	0.130*** (0.026)	0.163*** (0.034)	0.335*** (0.075)
Firm's processing status $_{i,t-1}$	5.771*** (0.713)	6.338*** (0.801)	6.356*** (0.780)
Number of observations	925916	521937	560222
Hansen statistic (p-value)	13.4 (0.77)	14.6 (0.63)	14.2 (0.22)
Marginal effects (dy/dx) of :	Measures of Financial constraints		
$\log(\text{Exchange rate})_{j,t}$ on $\log(\text{Export volumes})_{i,t}$	(1)	(2)	(3)
evaluated at:			
(1) Access to finance $_{i,t}$ (25th percentile)	1.647*** (0.304)	2.868*** (0.333)	2.178*** (0.468)
(2) Access to finance $_{i,t}$ (75th percentile)	1.426*** (0.213)	1.026** (0.459)	1.639*** (0.194)

Notes: The dependent variable is the (log) of firm's exports at constant prices. Exchange rate is the (log) of real effective exchange rate computed at the industry level; j is the industry to which firm i belongs. Financial constraints are approximated in different ways in columns (1) through (3) using alternative indicators of firm i 's ability to have access to external finance. Estimates are obtained using the system GMM panel estimator. The firm's processing status is defined by a dummy variable equal to one if the firm only engages in processing transactions. In each year, the sample comprises only the firms engaged in trade. Firms fixed effects are included. Firm size class dummies are constructed based on the year-specific deciles of the distribution of employment. We also include year and firm type dummies. Robust standard errors corrected for clustering are reported in parentheses. To deal with the possible endogeneity of the financial indicator, the instrument set of the GMM type includes lagged values of it dated $t-1$ and earlier. Hansen is a test of over-identifying restrictions and is asymptotically distributed as a χ^2 . Sample period: 2000-2006. *** denotes significance at the 1% confidence level; ** at the 5% confidence level and * at the 10% level.

sensitive to exchange rate than those undertaken by firms with a higher access to external finance. This finding is detected at both the intensive and extensive margins of exports. We delve into the mechanism underlying this result and provide evidence that the degree of exchange rate pass-through to export prices denominated in the domestic currency is indeed lower for the firms facing more financial constraints. This is consistent with the view that only firms with enough financial strength are able to pursue pricing-to-market strategies and absorb in their markups any exchange rate shock.

Moreover, we control for a variety of other firm level factors, apart from financial constraints, which may also contribute to shape the response of exports to exchange rate. These are the intensity of use of imported inputs, labor productivity, price stickiness and employment size. Each of this channel is found to play a role on empirical grounds and we therefore do not rule out their relevance. However, we show that, even controlling for these parallel channels, the heterogeneity across firms in their ability to have access to finance significantly affects the size of the reaction of exports to exchange rate. Furthermore, we establish that our main findings continue to hold when we allow for possible asymmetric effects of appreciation vs. depreciation of the exchange rate, when we employ firm-level exchange rates and when we control for the firm's processing status.

Although in this paper we focus solely on microeconomic data, we believe, however, that the characterization of firm's heterogeneity uncovered here for the exchange rate elasticity of exports is relevant for the understanding of the low values of this elasticity typically detected with aggregate data.

Appendix A

Let us show that there is an interval of values of the fixed cost of exporting, F_i , such that the impact of an exchange rate depreciation on the minimum required productivity is stronger for firms that face higher costs of external financing, ϕ_i . This requires studying the derivative of $\frac{\partial a_i^*}{\partial e}$ in Eq. (11) with respect to ϕ_i , that is:

$$\frac{\partial^2 a_i^*}{\partial e \partial \phi_i} = \frac{-1/2[ef(1+\phi_i)F_i - ef\phi_i L_i]^{-(1/2)}(fF_i - fL_i)\tau c(1+\phi_i) - \tau c(d - e^{-(1/2)})\sqrt{f(1+\phi_i)F_i - f\phi_i L_i}}{[\tau c(1+\phi_i)]^2}.$$

The condition for this expression to be positive is the following:

$$f[(1+\phi_i)F_i - \phi_i L_i] > -fL_i + 2d\sqrt{ef(1+\phi_i)F_i - ef\phi_i L_i}. \quad (16)$$

Similarly to expression (10), this condition, to be defined, requires that the argument of the square root in the right hand side is non-negative, that is $F_i \geq \frac{\phi_i}{1+\phi_i} L_i$. Consider the case when this condition is satisfied as an equality: $F_i = \frac{\phi_i}{1+\phi_i} L_i$. In this case, the left hand side of the expression above is zero, while the right hand side is equal to $-fL_i < 0$. The condition for the above expression to be positive is therefore satisfied. For small increases of F_i , leaving L_i unchanged, the condition will also be satisfied, which proves our statement.

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